

BEGINNER'S GUIDE: YOUR FIRST AUTONOMOUS AI

Step-by-Step Friendly Introduction to Building Intelligent Autonomous AI Bots with Sapphire

Version 1.0.0 (Beginner Edition) | Sapphire Language Architecture Team

1. Welcome! What is an Autonomous AI Agent?

Welcome to the exciting world of **Autonomous AI Programming** with Sapphire! If you have never built an AI before, do not worry. An **Autonomous AI Agent** is simply a computer program that can observe what is happening on your computer, think about what needs to be done using AI, and take action automatically.

Analogy: Chatbot vs Autonomous Agent

- **Chatbot:** Like a friend who only replies when you send a text.
- **Autonomous Agent:** Like a dedicated assistant who monitors your computer, fixes problems, trains smart models, and works 24/7 in the background without needing to be asked!

2. The 3 Steps of Every Autonomous Bot

Every autonomous bot in Sapphire follows 3 simple steps:

Step 1: Check the System (Perception) — The bot reads system metrics like RAM, CPU, or files using `os.system_info()`.

Step 2: Ask the AI (Intelligence) — The bot sends system data to `ai.prompt(...)` for decision making.

Step 3: Perform Action (Execution) — The bot posts desktop alerts with `os.notify()` or executes commands.

3. Writing Your Very First 5-Line AI Agent

Here is the simplest complete Autonomous AI Agent written in Sapphire (.sp):

```
// 5-Line Autonomous AI Agent in Sapphire (.sp)
let info = os.system_info();
let question = "System CPU is at {info.cpu_usage_percent}%. Recommend optimization action.";
let ai_opinion = ai.prompt(question);

os.notify("Sapphire Agent Alert", ai_opinion);
print("■ Agent Output: {ai_opinion}");
```

4. Line-by-Line Code Breakdown

Code Line	What It Does (Beginner Friendly)
<code>let info = os.system_info();</code>	Gets live CPU, RAM, and disk usage from your computer.
<code>let question = "...";</code>	Constructs a prompt string with your CPU percentage.
<code>let ai_opinion = ai.prompt(question);</code>	Passes the question to Sapphire's native AI engine.
<code>os.notify("...");</code>	Displays a pop-up toast notification on your Windows desktop!
<code>print("...");</code>	Prints the AI's recommendations to your screen.

5. How to Run Your Agent

You can run your Sapphire AI agent in 3 easy ways:

- Option A (Emerald Developer Studio):** Double-click Emerald_Studio.exe, select the 'Auto Agent' template, and click  Run.
- Option B (Command Line):** Open terminal and type `sapphire run bot.sp`.
- Option C (Voice Tutor):** Double-click sapphire_voice_tutor.exe for audio guided practice.

Next Steps & Challenges

Try customizing the prompt to inspect RAM load, or use ``scheduler.interval(60.0, fn)`` to make your bot monitor your PC every minute automatically!